AdaptEngine

Adaptive Learning, Diagnostic Assessment and Personalized Path-Planning Engine

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https://github.com/vhemanth06/AdaptEngine

Problem and Motivation

The problem

Online courses give every learner the same fixed order.

- Beginner with 10 min/day $\Rightarrow$ overwhelmed
- Experienced learner $\Rightarrow$ wastes time on known topics
- Deadlines and free time vary widely

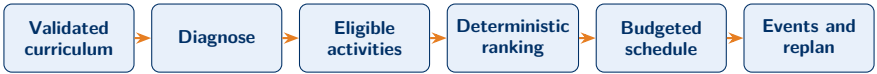
Goal

A personalization engine that finds **what the learner knows**, plans **what to study next** within goal, deadline and daily budget, and explains **why**.

Why it is hard

- Skill must be estimated from a very short diagnostic, with uncertainty
- Prerequisites are strict rules; time and difficulty are objectives
- Lessons, remediation and review compete for the same few minutes
- After a missed day or a changed deadline, the new plan must stay valid and keep completed work
- Every decision must be traceable so the learner can trust the plan

Proposed Approach



M1 – Curriculum

Prerequisite DAG of concepts, lessons and exercises with integer durations

M2 / M4 – Learner model

Adaptive diagnostic, BKT-style mastery with uncertainty, difficulty policy

M3 – Planner

Day-by-day plan for 5/10/15/30 min budgets, or a clear INFEASIBLE

M5 – Review

Spaced review by forgetting risk, plus remediation

M6 – Replanning

Missed day, failed test, deadline or budget change

M7 / M8 – Trust and proof

Decision traces; evaluation against fixed, static and greedy baselines

Key Design Decision

Hard constraints: rules every plan must satisfy

- Prerequisites require evidence-based mastery
- Daily study time must stay within B minutes
- If no valid plan meets the deadline, report INFEASIBLE with the shortfall

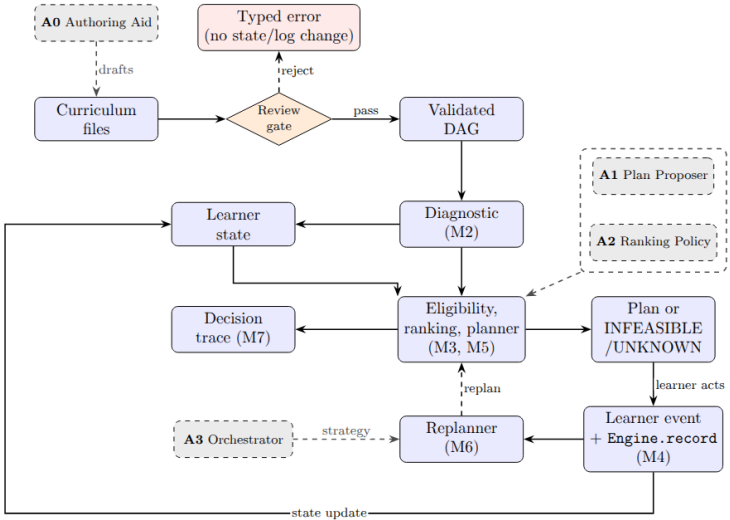
Soft objectives: choose what matters most

- Give higher priority to urgent reviews and activities that unlock more topics
- Prefer exercises that match the learner's current level
- Use these scores to order valid activities; they never override prerequisites or time limits

$$\text{score} = w_r \cdot \text{review urgency} + w_u \cdot \text{unlock count} + w_d \cdot \text{difficulty fit}$$

Ranks eligible activities; higher scores are scheduled first. Ties use activity ID.

System Design:Workflow



M1: Curriculum Representation

Prerequisite graph

Represent concepts and their dependencies as a directed acyclic graph (DAG). Each concept contains lessons and exercises with integer-minute durations.

Validation

- Check the curriculum schema
- Reject unknown prerequisite references
- Detect cycles before planning

Content integrity

- Review content before it is used
- Record its SHA-256 hash in a manifest
- Verify the hash at live entry points

M4: Mastery Model

Difficulty-aware Bayesian update

$$g(d) = 0.30(1-d), \qquad s(d) = 0.05 + 0.25d$$

$$P_c = p(1-s) + (1-p)g$$

$$p'_{\text{correct}} = \frac{p(1-s)}{p(1-s) + (1-p)g}, \qquad p'_{\text{wrong}} = \frac{ps}{ps + (1-p)(1-g)}$$

$$p'' = p' + (1-p')T \qquad (T_{\text{diag}} = 0,\ T_{\text{exercise}} = 0.05,\ T_{\text{lesson}} = 0.20)$$

Worked check: $p = 0.5$, $d = 0.5$, wrong answer

Bayesian posterior: $p' \approx 0.1707 \longrightarrow$ After exercise transition: $p'' \approx 0.2122$

Parameters are synthetic heuristics, not fitted values.

M2: Adaptive Diagnostic

Choose the item expected to reduce uncertainty most

$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$$

$$\mathrm{EIG}(a) = H(p) - \left[P_c H(p'_{\text{correct}}) + (1 - P_c) H(p'_{\text{wrong}}) \right], \qquad a^* = \arg\max_a \mathrm{EIG}(a)$$

Diagnostic policy

- Visit concepts in stable topological order
- At most 8 items total and 3 per concept
- Stop when $H(p) \leq 0.50$ or the item cap is reached
- Cold start: $p_0 = 0.20$, `UNASSESSED`
- If $p < 0.40$, skip descendant concepts
- Deterministic tie-breaks; record the stop reason

M3: Planner and Feasibility Check

How a plan is built

1. **Needed work:** walk back from the goal, drop concepts already mastered with evidence, project lessons and exercises still required
2. **Cheap bounds:** total work, longest dependency chain and longest activity vs. deadline $\times$ budget
3. **Pack each day:** only prerequisite-ready activities that fit; scores order them; a non-fitting one is deferred
4. **Exact search:** branch-and-bound packing when greedy overruns the deadline

Difficulty policy

Pick the exercise with $P(\text{correct})$ closest to **75%**; ties by $|d - 0.5|$, then ID.

Three outputs

FEASIBLE: valid day-by-day plan

INFEASIBLE: proven, with shortfall (days, minutes) and options

UNKNOWN: node limit reached; never a false INFEASIBLE

Why exact search matters

Items 8, 8, 4 min, budget 10, 2 days: total 20 passes the cheap checks, yet no 2-day packing exists. Exact search proves INFEASIBLE.

M5–M7: Review, Replanning and Traces



Spaced review and remediation

- Recall $r(t) = 2^{-\Delta t/h}$, $h = 30$ days; review due when $r < 0.80$
- Urgency $= (1 - r)(1 + \text{descendants})$
- Failed answer with $p < 0.40 \Rightarrow$ repeat lesson first; flag cleared after it
- Untouched concepts are not mistaken for failed ones

M7: Decision traces

Every selected, omitted and deferred activity records its reason and score parts; saved as JSON.

M6: Replanning is a fresh run from today

Missed day	zero minutes; replan from next day
Failed test	mastery down; remediation at top
Deadline change	new plan or INFEASIBLE proof
Budget change	repack remaining work

Completed work is preserved, never lost or repeated.

Project Progress

Milestone	Component	Progress	Status
M1	Curriculum	Validated curriculum DAG and content integrity checks	Done
M2	Diagnostic	Adaptive selection and stopping; error vs. baselines still open	Done
M3	Planner	Eligibility, ranking, budgeted schedule, and feasibility oracle	Done
M4	Mastery model	Bayesian updates, real pyBKT adapter, difficulty policy	Done
M5	Review	Spaced review and remediation priorities	Done
M6	Replanning	Replan after learner events or changed constraints	Done
M7	Decision traces	Explain activity selection, omission, and rescheduling	Partial
M8	Evaluation	Paired baseline run and TC8.1 done; TC8.2 (replanning vs. recomputation) planned for Week 6	Partial

Limitations and Next Steps

Limitations

- Prerequisites form a correct DAG
- One concept and one fixed difficulty per exercise; answers are right or wrong
- Activities are indivisible; no partial credit or hints
- BKT and forgetting parameters are heuristic, with one constant half-life

Next steps

- **W6:** TC8.2 replanning vs. recomputation; tune stopping threshold and item cap to cut diagnostic error
- **W7:** larger and more diverse cohorts; sensitivity to θ, k, half-life h
- **W8:** reproducible demo, final evaluation report, documentation

Thank you

Questions and discussion

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